Interaction in Visualization

Understanding interaction through real cases

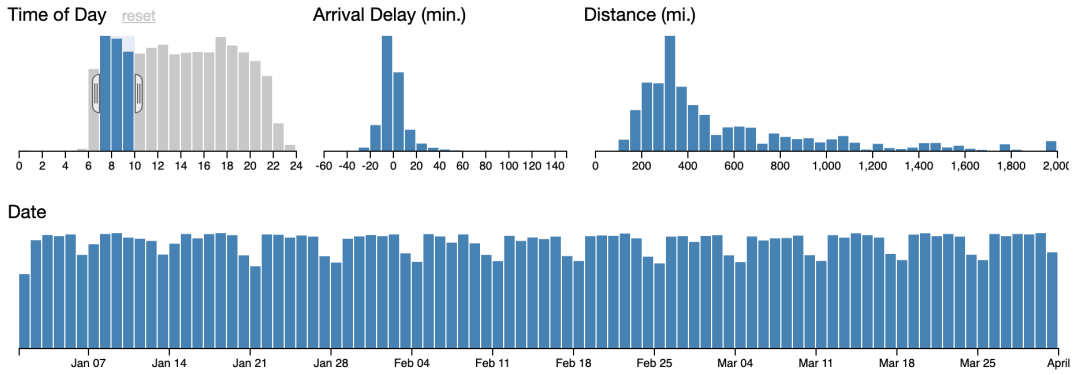
Information Visualization · Lecture 12 · Week 7

Five levels of interaction design

01	**Framework**	How does an action advance an analytical question?
02	**Operations**	What can the user change?
03	**Interaction in Multiple-View Visualizations**	How do operations and views work together?
04	**Design principles**	What makes the interaction understandable and usable?
05	**Evaluation**	Does the design help with the task, and at what cost?

LEVEL 01 · FRAMEWORK

What changes when we select morning flights?



March 31, 2001

46,331 of 231,083 flights selected.

09:59 AM	PHX	OAK	646 mi.	-4 min.
09:55 AM	MCI	SEA	1,489 mi.	+39 min.
09:55 AM	LAS	BNA	1,588 mi.	+3 min.
09:55 AM	GEG	OAK	723 mi.	+2 min.
09:55 AM	LAX	SLC	590 mi.	+0 min.

Square / Crossfilter · historical airline sample

Open case ↗

Start with a question

Do arrival delays look different for morning flights?

Demonstrate: clear the Date filter, then brush 07:00–10:00 on Time of Day.

Observe: the selected range, flight count, other distributions and rows.

What would you investigate next?

LEVEL 01 · FRAMEWORK

The complete interaction loop

Interaction is an iterative process in which a user acts on a visualization, the system changes its state and presents a response, and the user interprets that response to decide what to do next.

1 · Intent	2 · Action	3 · Operation	4 · Response	5 · New question
Compare morning delays	Drag across 07:00–10:00	Select a range; filter linked data	Counts, charts and rows update	Does distance explain the pattern?

The new question starts another cycle. Visible state records the current choices.

Framework adapted from Heer & Shneiderman, Interactive Dynamics for Visual Analysis (2012).

LEVEL 02 · OPERATIONS

Three intention groups, seven interaction operations

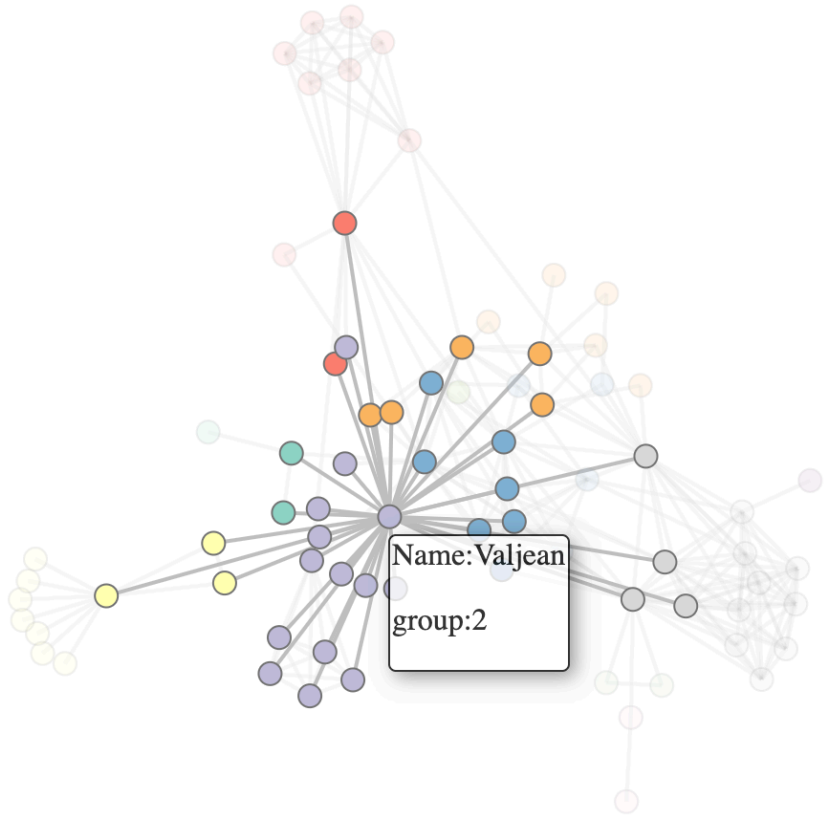
We classify interaction by the **analytical intention** it serves and the **state the user changes**.

Analytical intention	Interaction operations	What changes
Focus on objects	Select · Filter	Working set · eligible records
See the data differently	Navigate · Reconfigure · Encode · Abstract / elaborate	Viewpoint · layout · mapping · detail
Relate evidence	Connect	Visible relationships across objects or views

These three groups are not mutually exclusive. One analytical task can span several intentions and combine operations.

Operation types adapted from Yi et al. (2007). The three intention groups organize this lecture.
Navigation covers viewpoint changes within their broader Explore category.

Selection identifies a working set



Xianbin Gu · Les Misérables / D3

Open case ↗

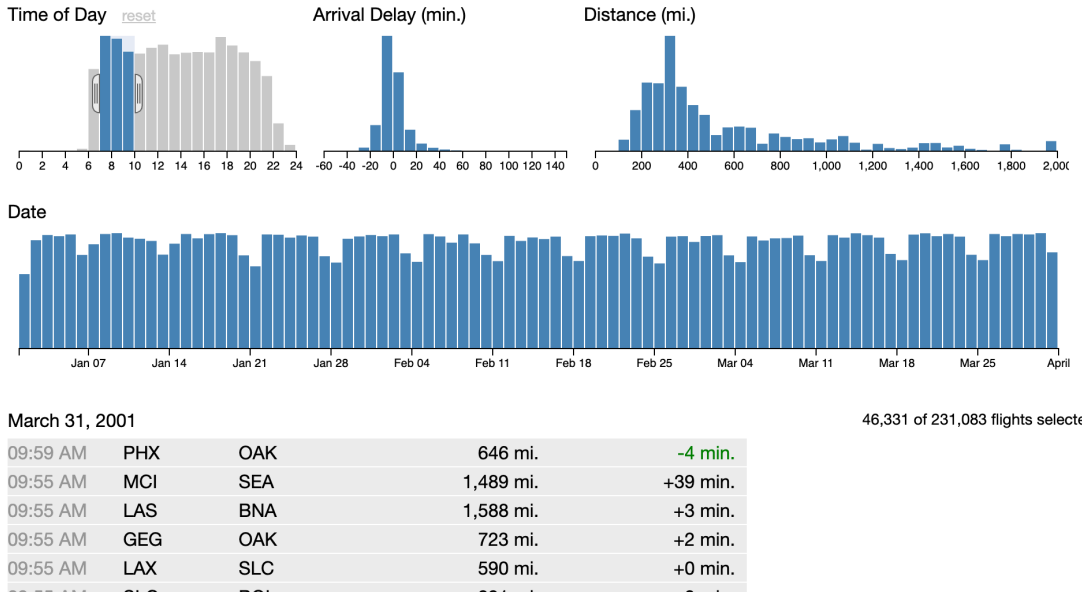
Selection identifies items as the current focus or targets for further operations.

Demonstrate: hover Valjean, then move away.

State change: the working set goes from no focused character to Valjean, then clears.

The tooltip and highlighting make this temporary focus visible.

Filtering changes the eligible records



Square / Crossfilter · historical airline sample

Open case ↗

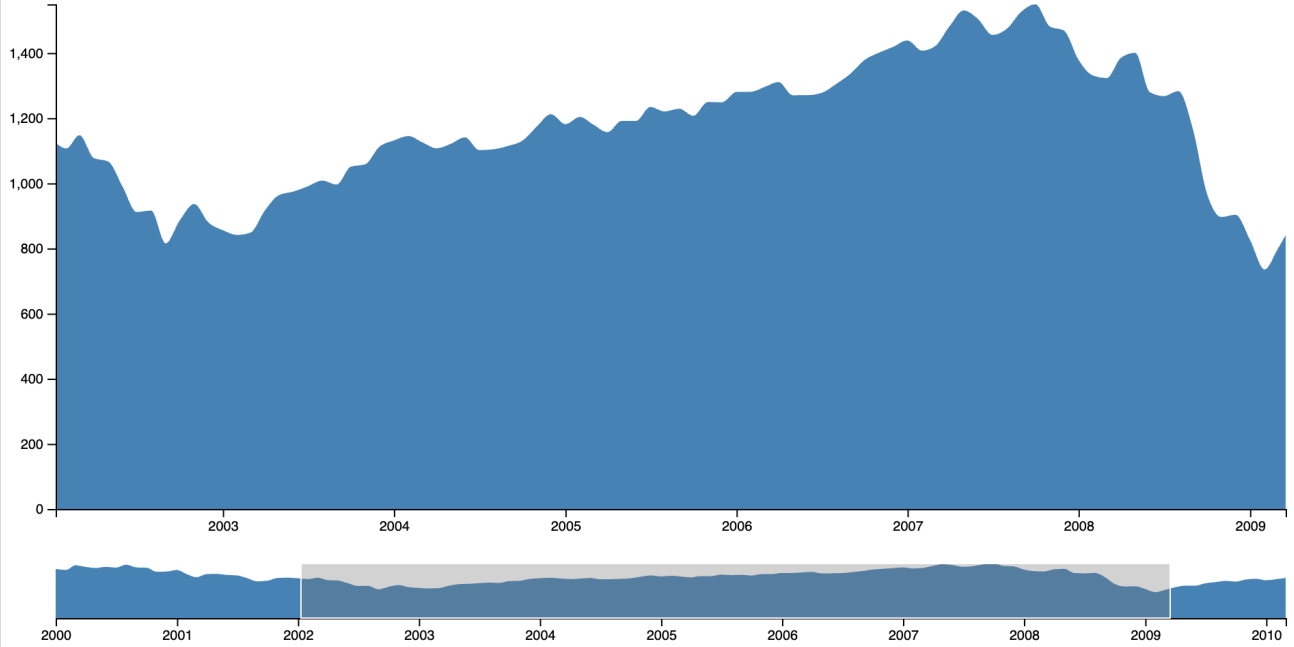
Filtering restricts which records contribute to a target view or calculation.

Demonstrate: clear filters, then brush 07:00–10:00.

State change: eligible records change from all flights to those matching the time interval.

The count, linked charts and flight list update. Gray bars retain the full distribution as context.

Navigation changes the viewpoint



Xianbin Gu · D3 brush / zoom · historical S&P 500

Open case ↗

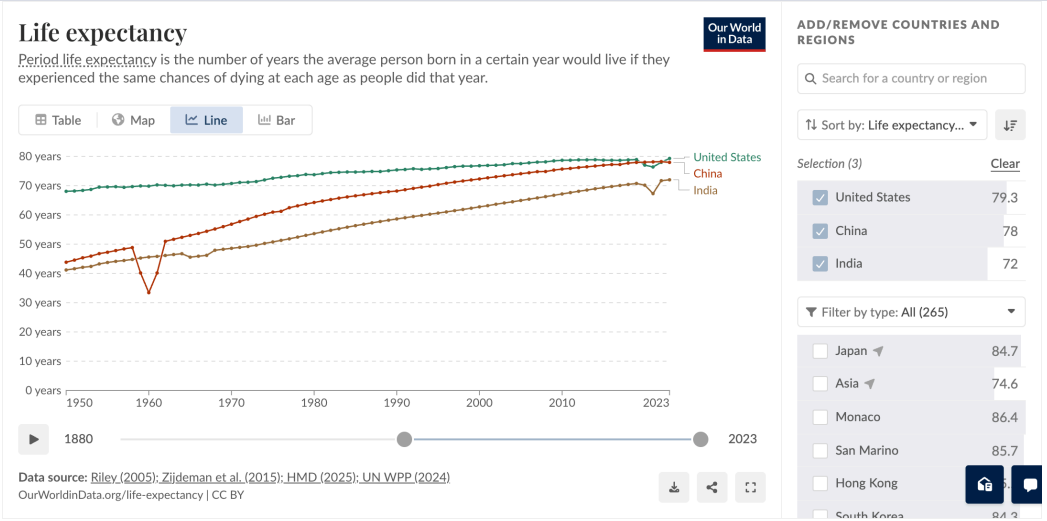
Navigation changes the visible region or scale of a data view.

Demonstrate: zoom over the upper chart, then drag to pan.

State change: the visible time interval and x-axis scale change.

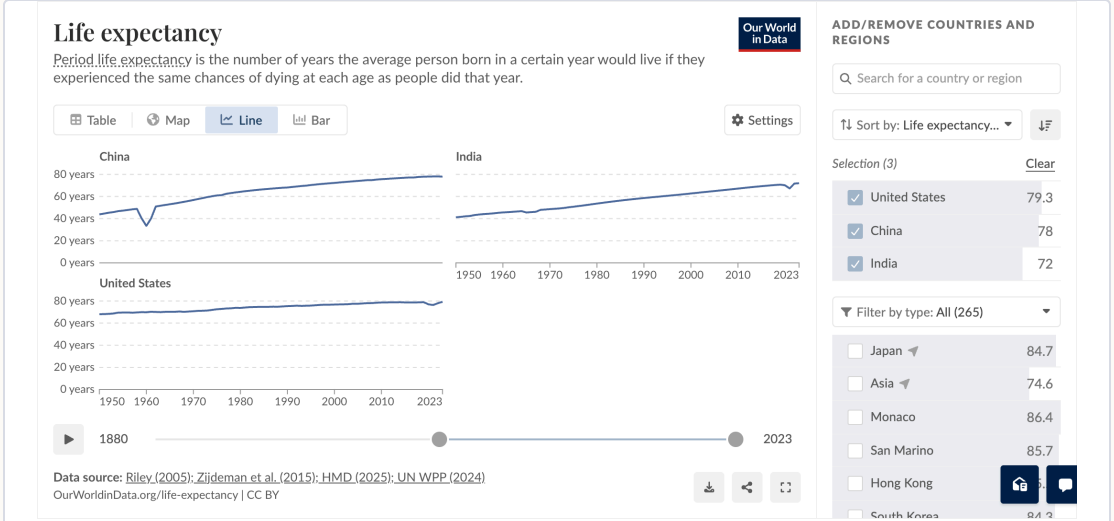
The underlying series stays the same. The lower brush marks the visible interval.

Reconfiguration changes the layout



OWID · CC BY · Overlaid layout

Open overlaid view ↗



OWID · CC BY · Split layout

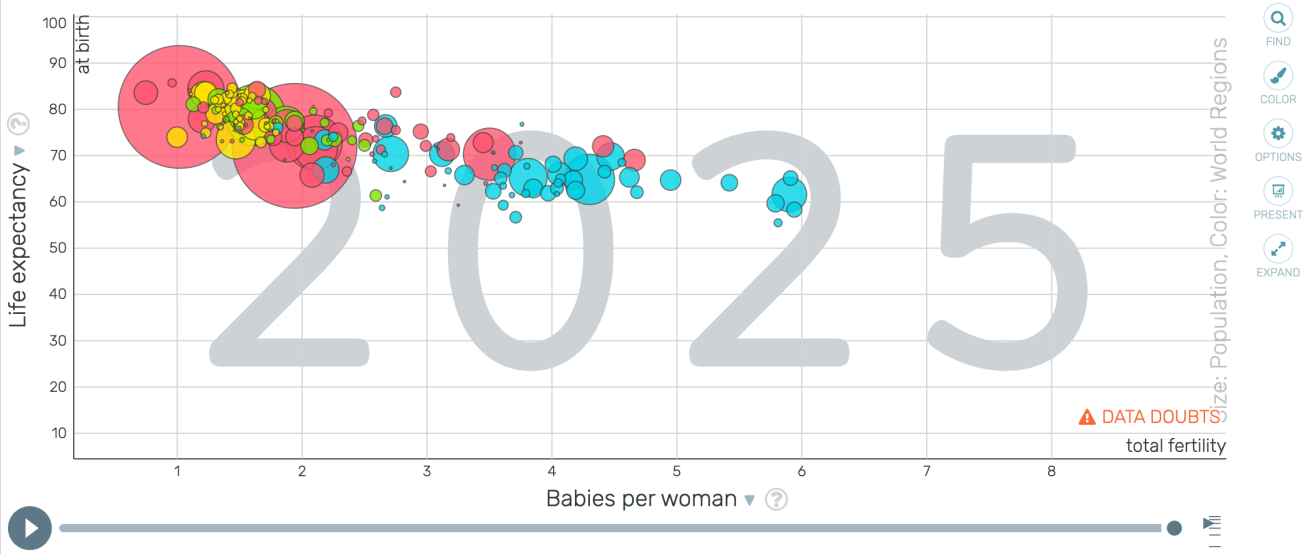
Open split view ↗

Reconfiguration changes the arrangement of an existing representation.

Demonstrate: switch between the two linked OWID layouts.

State change: three overlaid country lines move into separate panels. Countries, time range and measures stay fixed.

Encoding changes what a channel means



Gapminder Foundation · Gapminder Tools

Open case ↗

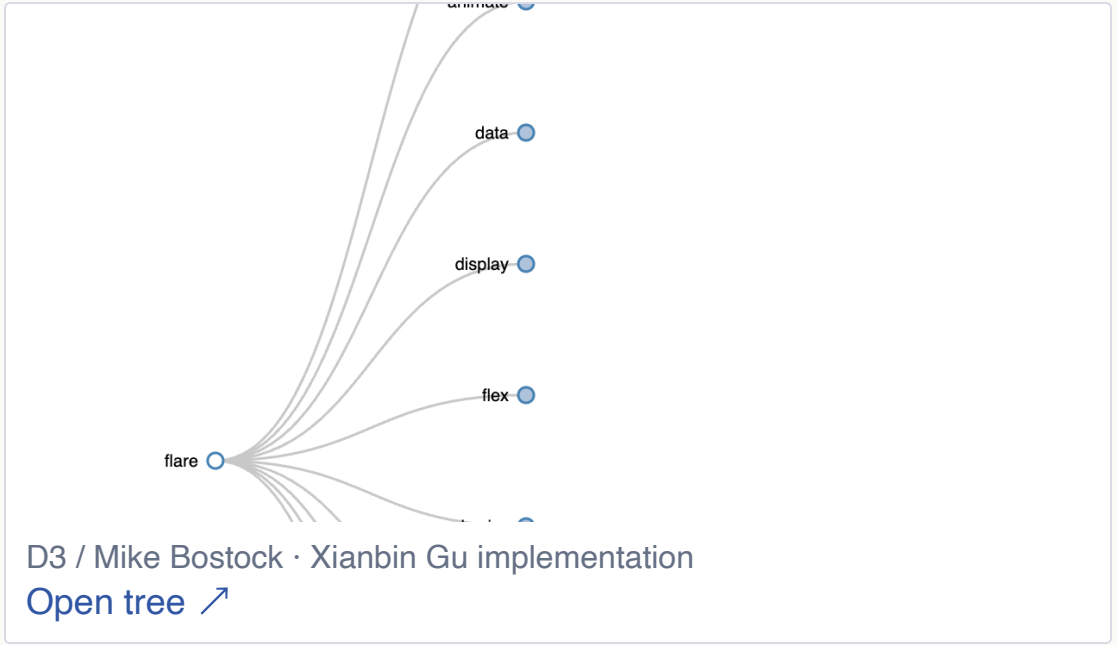
Encoding changes how data attributes map to visual marks or channels.

Demonstrate: change the x-variable from GDP per capita to Babies per woman.

State change: the mapping of x-position changes from GDP per capita to fertility.

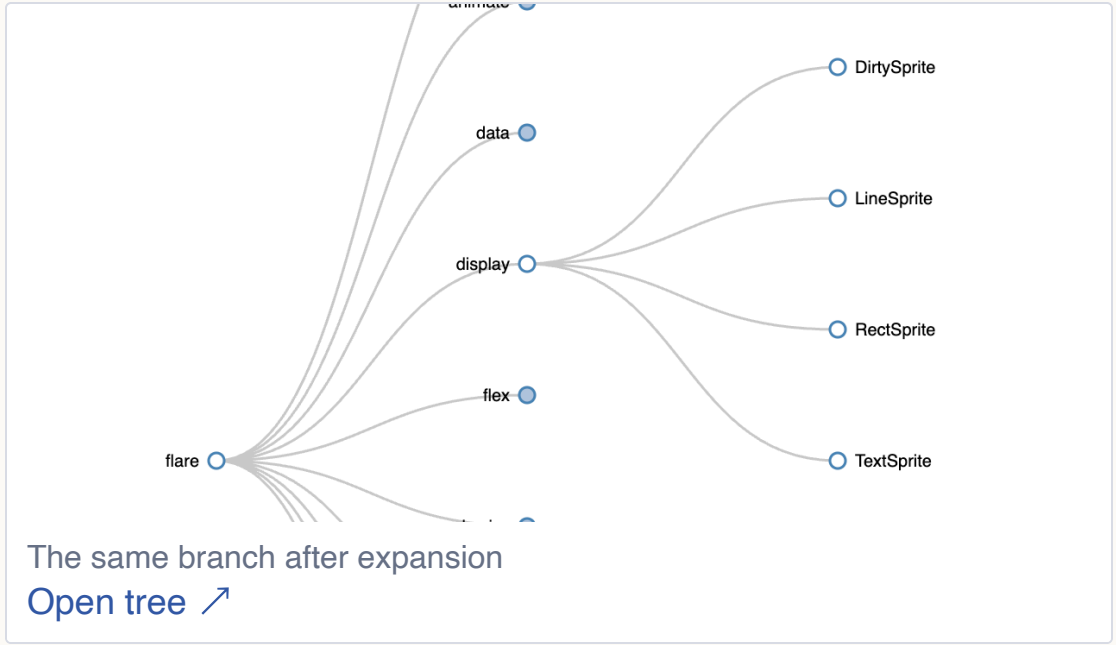
Country positions and the x-axis update. Keep the year and y-variable fixed.

Abstraction and elaboration change detail



D3 / Mike Bostock · Xianbin Gu implementation

Open tree ↗



The same branch after expansion

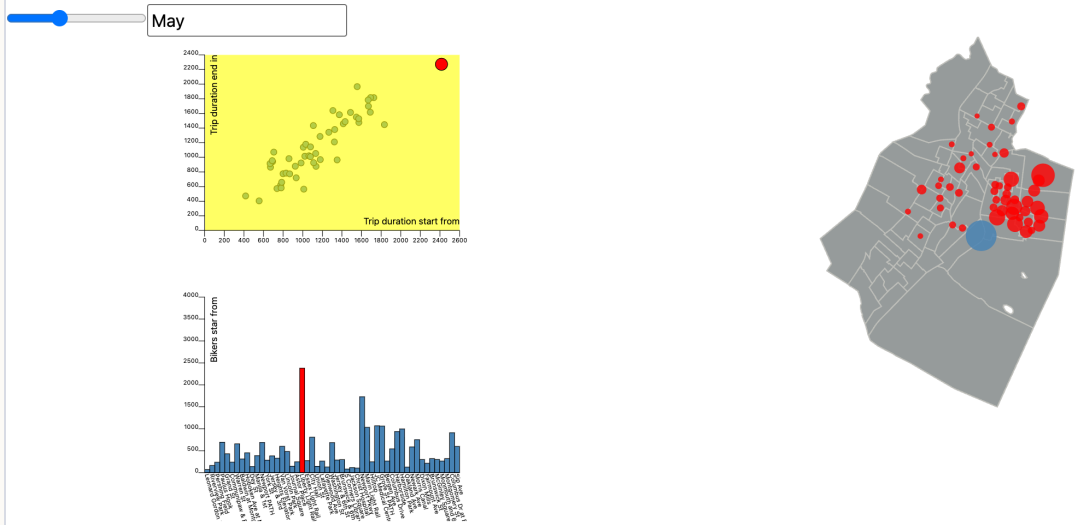
Open tree ↗

Abstract / elaborate reduces or increases the visible level of detail.

Demonstrate: click **display** to expand, then click again to collapse.

State change: display’s four child nodes and links appear or disappear. The stored hierarchy stays the same.

Connection makes relationships visible



Xianbin Gu · Citi Bike coordinated dashboard

Open case ↗

Connection reveals relationships between items or their representations.

Demonstrate: hover a station on the map.

State change: the hovered station's map mark, point and bar become emphasized together.

The shared station ID connects these marks. The data values stay the same.

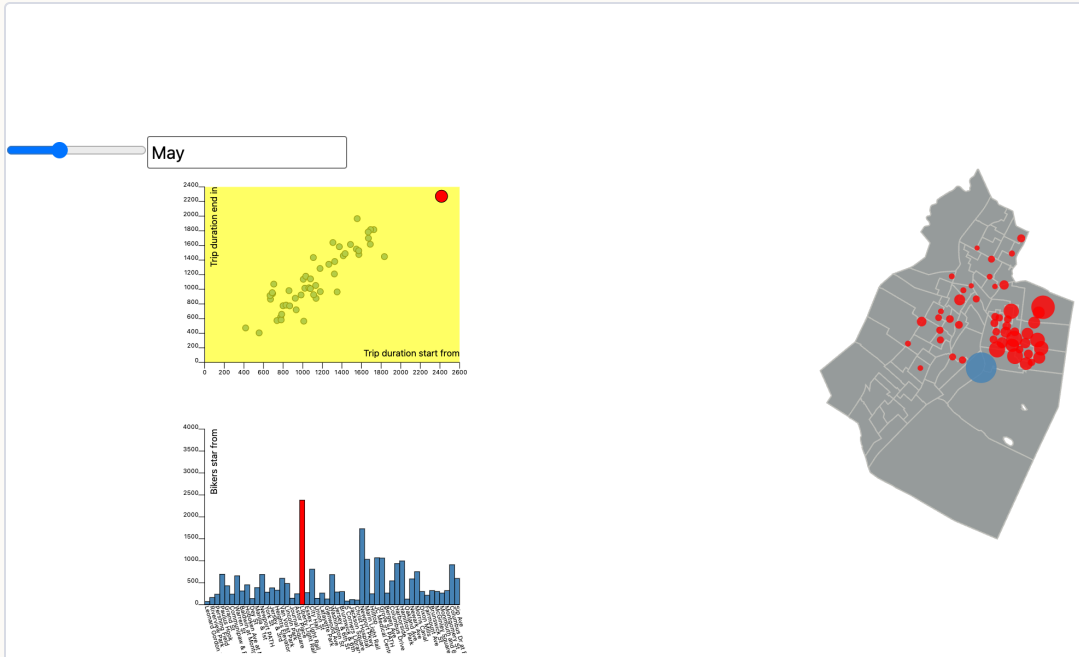
Interaction in Multiple-View Visualizations

A **view-combination pattern** specifies how several views share data, context or state to support a task.

View-combination pattern	Involved operations
Coordinated multiform views	Select + Connect (e.g., Citi Bike)
Overview + detail	Select + Navigate + Connect (e.g., S&P 500)
Small multiples	Filter + Reconfigure (e.g., OWID)

Operations work together through shared state or a sequence of choices. These are case-specific combinations. A static layout alone is not interaction.

Multiform views coordinate different perspectives



Xianbin Gu · Citi Bike coordinated dashboard

Open case ↗

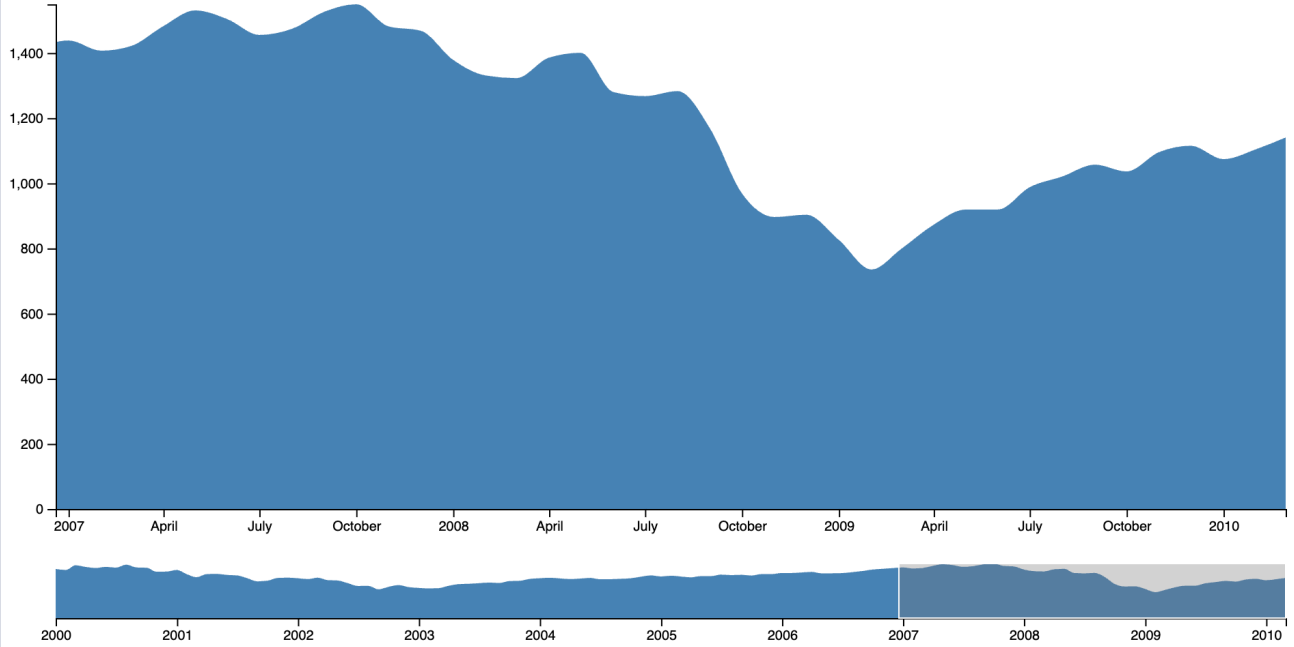
Multiform views show the same items using different representations. Coordination links their interaction state.

Task: locate a station and compare its attributes.

Select: hover a station on the map to set the focus.

Connect: its station ID links the highlighted point and bar to the map mark. Read its location and attributes together.

Overview + detail preserves context



Xianbin Gu · D3 brush / zoom · historical S&P 500

Open case ↗

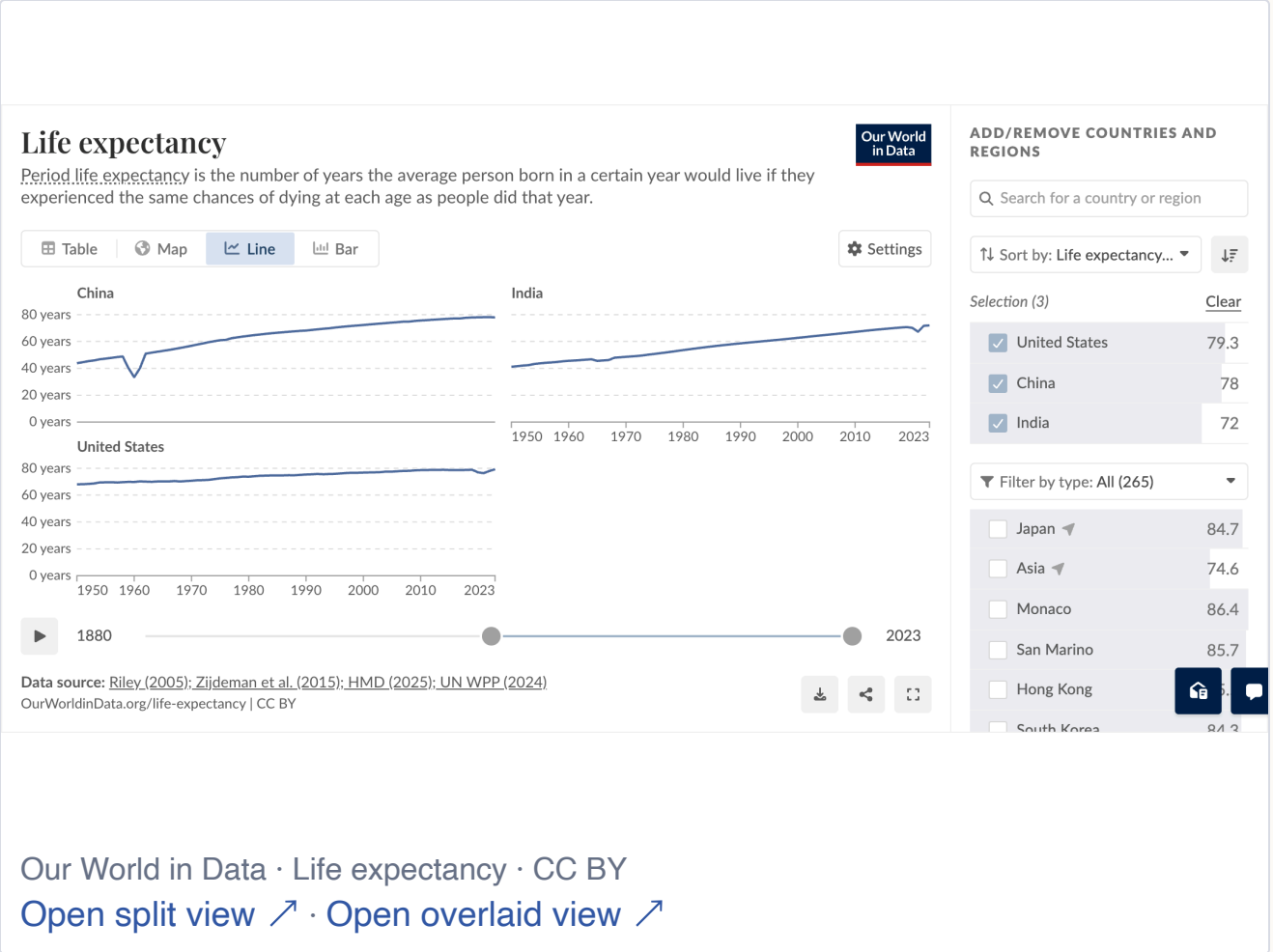
Overview + detail pairs a focused view with a broader view that preserves context.

Task: inspect a short period within the full history.

Select + Navigate: move or resize the lower brush. The selected interval sets the upper time window.

Connect: the overview brush stays synchronized with the detail window, showing where the enlarged interval belongs.

Small multiples preserve comparisons



Our World in Data · Life expectancy · CC BY

Open split view ↗ · Open overlaid view ↗

Small multiples repeat a common chart design for different subsets.

Task: compare country trajectories over the same years.

Filter: keep China, India and the US to define the comparison set.

Reconfigure: switch between overlaid lines and separate panels using the two links. Keep the countries, years and scales fixed to compare the same histories.

Make state changes match user expectations

Users should be able to **predict how an action changes interface state** and **understand the response**.

Crossfilter example: reset Time of Day while keeping the Date filter.

Stage	What the user should be able to do	Principles
Before acting	Predict that only the time constraint will be removed.	Clear mappings Consistency
After acting	Confirm: time brush clears, Date remains, linked views update.	Feedback Visible state
Next step	Compare morning flights with all hours for the same dates.	Recovery Meaningful control

Accessible interaction supports every stage. The goal is a useful analytical result.

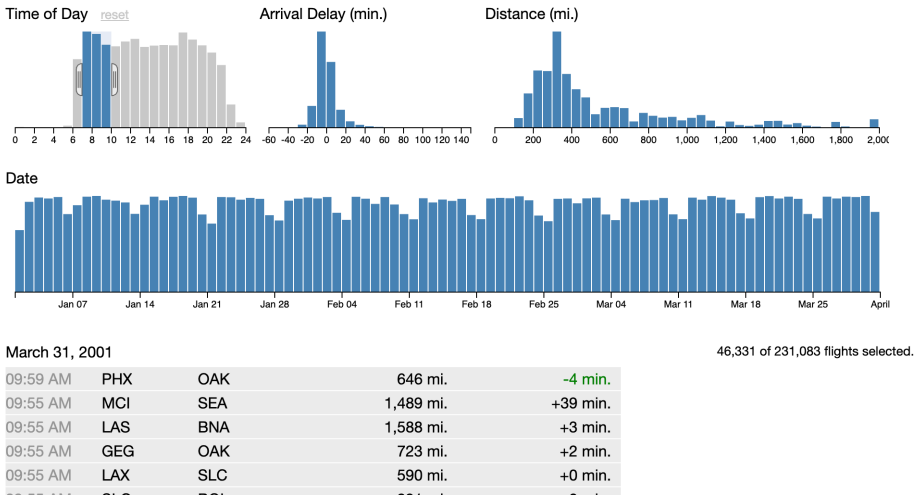
Case: Square / Crossfilter · Principles: Shneiderman et al., Designing the User Interface.

Principles make the interaction usable

Design principle	Involved operations / view patterns
Feedback	Select + Filter · Coordinated multiform views (e.g., Crossfilter)
Clear mappings	
Visible state	
Recovery	
Consistency	Select + Connect · Coordinated multiform views (e.g., Citi Bike)
Meaningful control	Filter + Reconfigure · Small multiples (e.g., OWID) Inspect Select / Filter / Reconfigure · Overview + detail analogy (e.g., Bloomberg)
Accessible interaction	Select + Connect · Proposed overview + detail (e.g., Les Misérables)

Principles: Shneiderman et al., Designing the User Interface; W3C WCAG 2.2.

Feedback and visible state explain the result



Square / Crossfilter · historical airline sample

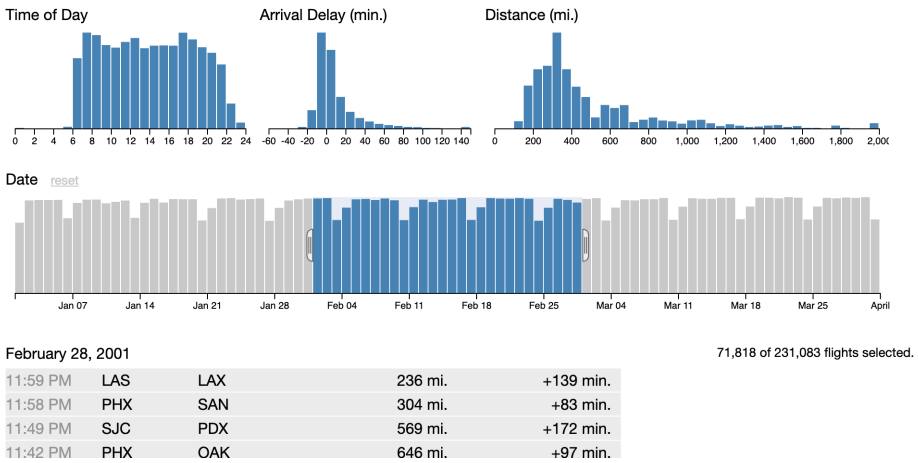
Open case ↗

Feedback communicates the response to an action. **Visible state** shows which choices remain active.

Improper design: hiding the active filter and providing unclear feedback can leave users unsure whether brushing took effect and make a subset look like all flights.

Proper design: after brushing 07:00–10:00, the count and linked charts update and the brush remains visible. Users can confirm the response, identify the active filter and interpret arrival delays for the intended flights.

Clear mappings make recovery predictable



Square / Crossfilter · historical airline sample

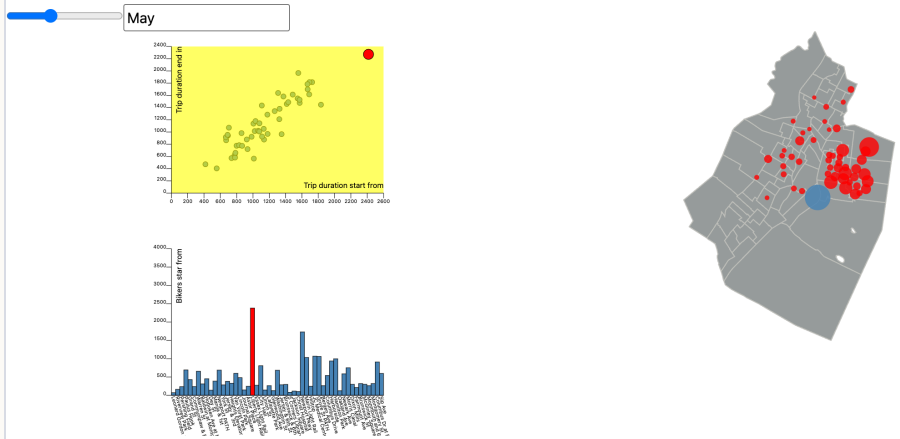
Open case ↗

Clear mappings make an action's effect and scope understandable. **Recovery** lets users restore an intended state after an action.

Improper design: an ambiguous reset can discard Date when users only want to clear Time of Day, forcing them to rebuild their comparison.

Proper design: each filter has its own adjacent reset. After Select + Filter, reset only Time of Day: Date stays active and coordinated views update, letting users compare morning flights with all hours within the same dates.

Consistency makes linked identity easier to follow



Xianbin Gu · Citi Bike coordinated dashboard

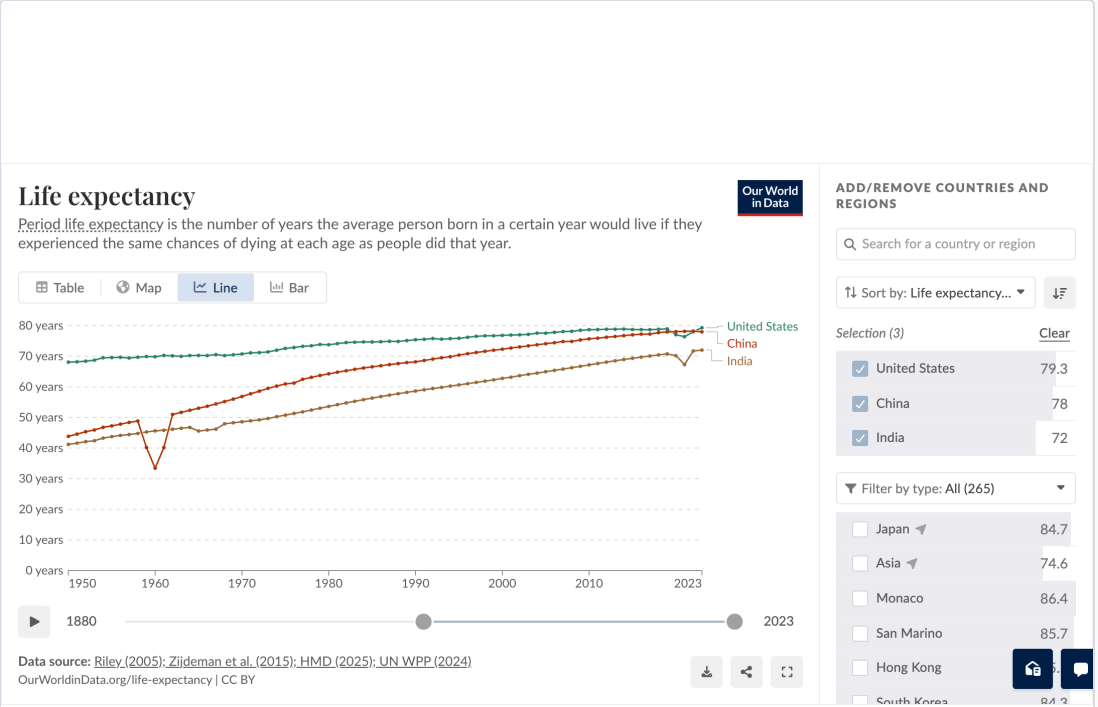
Open case ↗

Consistency keeps meanings and responses stable across related controls and views.

Improper design: using different emphasis cues for the same selected station can make its map mark, point and bar look unrelated across linked views.

Proper design: in a proposed redesign, hovering would Select a station and Connect its marks with a shared outline or label, preserving data colors. Users could track the same station and compare attributes across multiform views.

Meaningful control supports the analytical question



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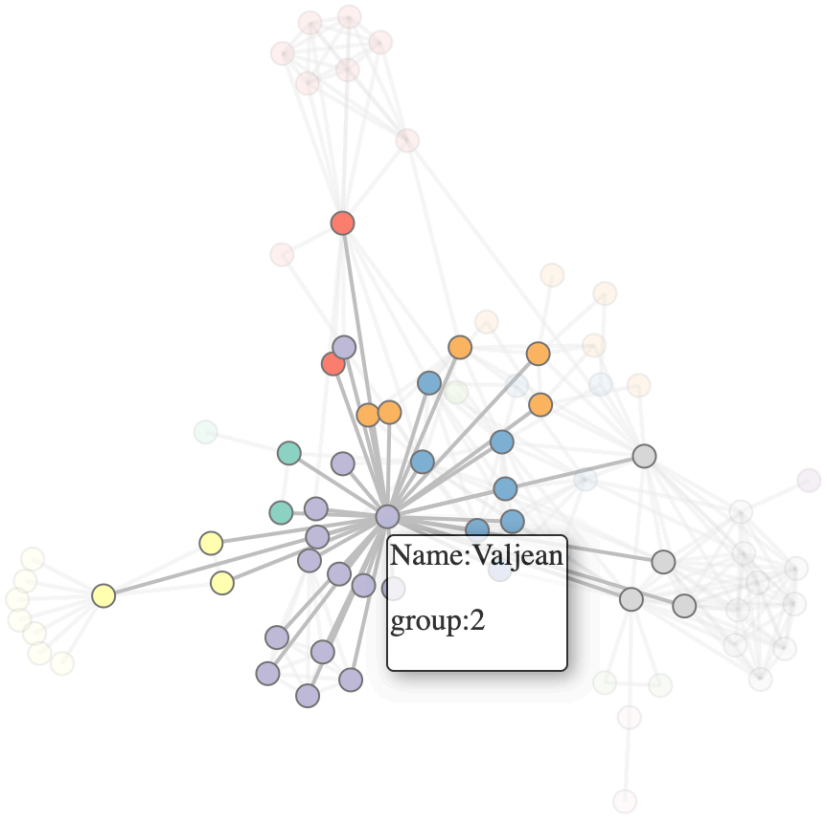
Open overlaid view ↗ · Open split view ↗

Meaningful control gives users choices about data and representation that serve their analytical question.

Improper design: fixing the country set or layout can prevent users from choosing relevant comparisons or following individual trajectories.

Proper design: users Filter to China, India and the US, then Reconfigure using the two layout links. Small multiples help trace each trajectory. The overlay helps compare country levels over the same years.

Accessible interaction supports the task



Xianbin Gu · Les Misérables / D3

Open case ↗

Accessible interaction provides ways to operate controls and perceive their state across different input and access needs.

Improper design: if revealing relationships depends only on precise pointing or sustained hover, some users may be unable to inspect neighbors. Keyboard access in this case still needs checking.

Proper design: a proposed redesign would let users Select a character by keyboard and Connect it to a persistent neighbor list. This overview + detail arrangement would keep relationships available without sustained hover.

Evaluation turns principles into observable questions

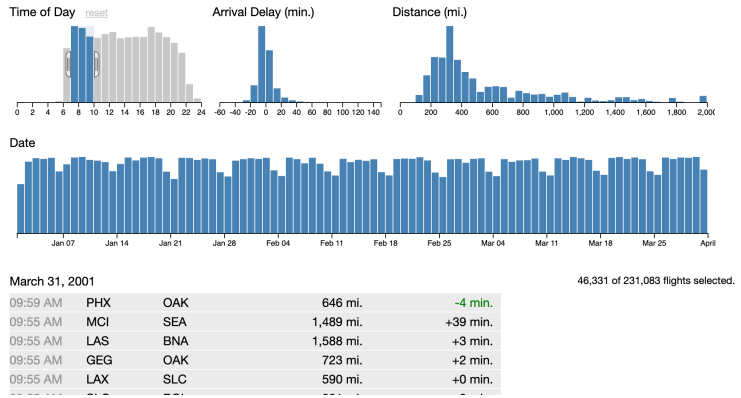
Interaction evaluation tests whether users can complete a specified analytical task accurately and with acceptable effort. **Design principles** state the requirements; evaluation checks them through observed task performance.

Evaluation question	Design principles
Does it help answer the question?	**Meaningful control**
What changed, and what remains active?	**Feedback / Visible state**
Can users predict what related controls do?	**Clear mappings / Consistency**
Can users restore the intended state?	**Recovery**
Can users operate and perceive it?	**Accessible interaction**

LEVEL 05 · EVALUATION

Crossfilter: audit a complete filtering task

Task: filter morning flights, inspect arrival delays, then reset.



Square / Crossfilter · historical airline sample

Open case ↗

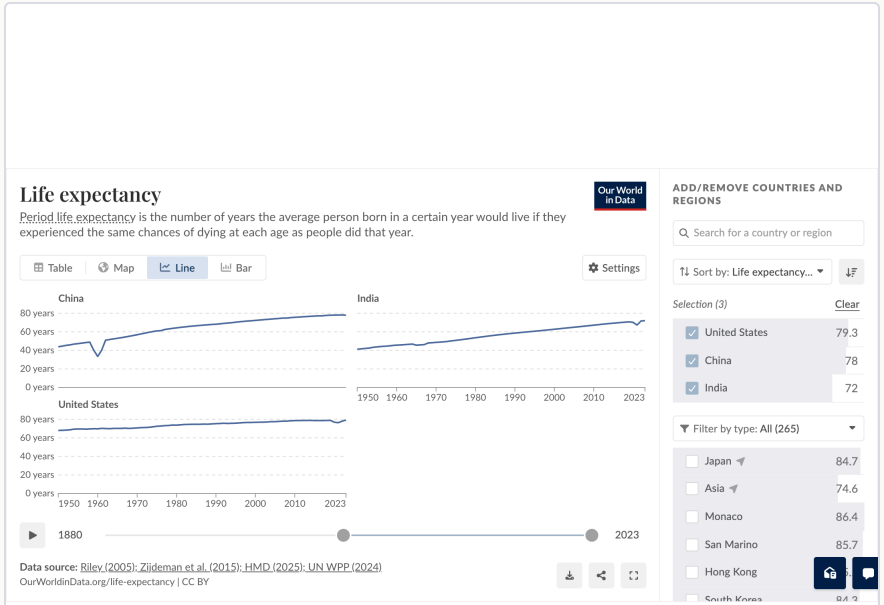
What changed, and what remains active? (Feedback / Visible state)
Check: brush morning hours; ask users to describe the response and active filters. **Evidence:** correct chart and scope descriptions.

Can users predict what related controls do? (Clear mappings / Consistency)
Check: predict each local reset's effect. **Evidence:** predictions match results across filters.

Can users restore the intended state? (Recovery)
Check: reset Time of Day with Date active. **Evidence:** Date remains; record unintended changes.

OWID: compare benefit with interaction cost

Task: trace one country's history, then compare China, India and the US in 2000.



Our World in Data · Life expectancy · CC BY

Open overlaid view ↗ · Open split view ↗

Does it help answer the question? (Meaningful control)

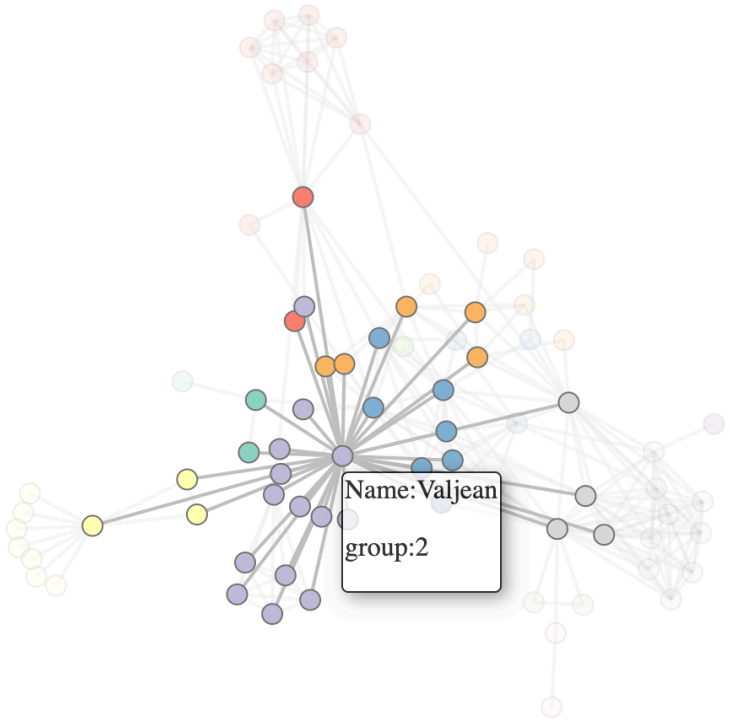
Check: perform both tasks in the overlaid and split layouts. Keep countries, years and scales fixed; verify answers against the data.

Evidence: compare answer accuracy, completion time, view switches and requests for help for each task.

Judge whether the layout supports a correct answer with acceptable effort. Faster interaction alone is not enough.

Network: current case and proposed prototype

Task to test: use only a keyboard to identify a character and its neighbors.



Current case · Xianbin Gu · Les Misérables / D3

Open case ↗

Current case: Can users operate and perceive it? (Accessible interaction)
Check: attempt the task. **Evidence:** correct names, perceivable focus/state, blocked steps and help needed.

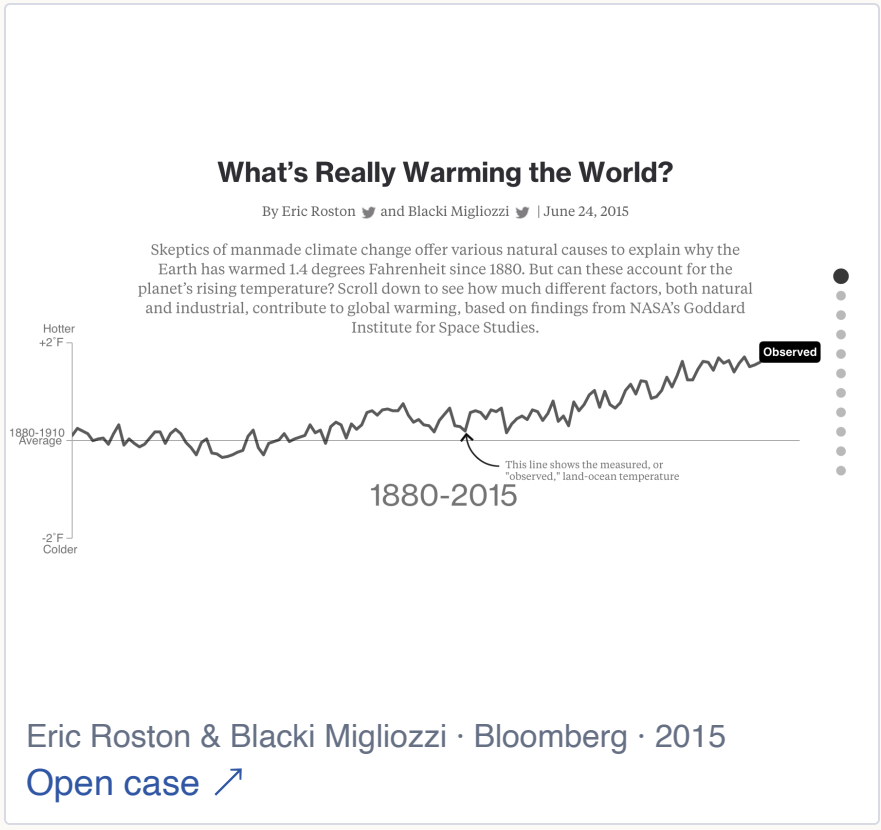
Proposed prototype: repeat the accessibility check with keyboard selection and a persistent neighbor list. Compare the same evidence.

Can users restore the intended state? (Recovery)
Check on the prototype: change or clear a mistaken selection. **Evidence:** correct final state, unintended changes and extra actions.

LEVEL 05 · EVALUATION

Bloomberg: evaluate the reader's control

Task: explain the argument, then revisit its evidence. **Reader agency** applies **Meaningful control** to a story's pace, path and analytical choices.



Eric Roston & Blacki Migliozzi · Bloomberg · 2015

Open case ↗

Does it help answer the question? (Meaningful control)
Check: return to a relevant comparison and explain how it supports the claim. **Evidence:** valid explanations, successful returns, navigation steps and help needed.

What changed, and what remains active? (Feedback / Visible state)
Check: move between stages; ask what changed and which factors remain shown. **Evidence:** correct identification of chart changes, the current stage and comparison.

Five levels connect one action to a useful outcome

1. **Framework:** connect a question to an action and a response.
2. **Operations:** identify what changes.
3. **Interaction in Multiple-View Visualizations:** organize views around a task.
4. **Principles:** make state and control understandable.
5. **Evaluation:** test whether the interaction helps.

Lecture 13: use these mechanisms to organize evidence and guide a data story.